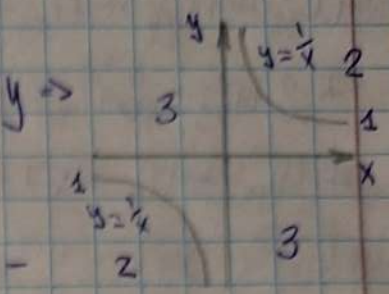
Первое задание.

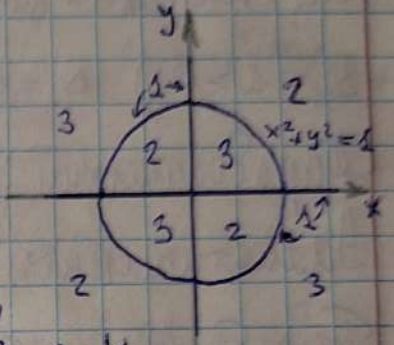
Т1.1.



а) $yu_{xx} + 2u_{xy} + xu_{yy} - u_y = 5x$. $D = 1 - xy \Rightarrow$

при $y = \frac{1}{x}$ - параболический тип (1);

при $y > \frac{1}{x}$ - эллиптический (2); при $y < \frac{1}{x}$ - гиперболический (3).



б) $(x^2+y^2-1)u_{xx} + xy u_{yy} - u_{xx} = 0$.

$D = -xy(x^2+y^2-1) \Rightarrow$ при $\left[\begin{array}{l} x=y=0 \\ x^2+y^2=1 \end{array}\right.$ — параболический (1); при $\left\{\begin{array}{l} xy>0 \\ x^2+y^2>1 \end{array}\right.$ и $\left\{\begin{array}{l} xy<0 \\ x^2+y^2<1 \end{array}\right.$ - эллиптический (2); при $\left\{\begin{array}{l} xy<0 \\ x^2+y^2>1 \end{array}\right.$ и $\left\{\begin{array}{l} xy>0 \\ x^2+y^2<1 \end{array}\right.$ - гиперболический (3).

№2.1(2).

$4u_{xx} - 4u_{xy} - 2u_{yz} + u_y + u_z = 0$

Уравнение характеристик:

$4(\partial_x)^2 - 4\partial_x\partial_y - 2\partial_y\partial_z = (2\partial_x - \partial_y)^2 - (\partial_y)^2 - 2\partial_y\partial_z = (2\partial_x - \partial_y)^2 - (\partial_y + \partial_z)^2 + (\partial_z)^2 \Rightarrow \left\{\begin{array}{l} \partial_1 = 2\partial_x - \partial_y \\ \partial_2 = \partial_y + \partial_z \\ \partial_3 = \partial_z \end{array}\right. \Rightarrow C = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

Преобр. координат: $T = (C^{-1})^T$. $C^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow$

$T = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -\frac{1}{2} & -1 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix} = T \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \left\{\begin{array}{l} \xi = \frac{1}{2}x \\ \eta = \frac{1}{2}x + y \\ \zeta = -\frac{1}{2}x - y + z \end{array}\right. \Rightarrow$

$u_x = \frac{1}{2}u_\xi + \frac{1}{2}u_\eta - \frac{1}{2}u_\zeta$; $u_{xx} = \frac{1}{2}\left(\frac{1}{2}(u_{\xi\xi} + u_{\eta\eta} + u_{\zeta\zeta}) + u_{\xi\eta} - u_{\xi\zeta} - u_{\eta\zeta}\right)$; $u_y = u_\eta - u_\zeta$; $u_z = u_\zeta$; $u_{xy} = \frac{1}{2}u_{\xi\eta} +$

$+\frac{1}{2}u_{\eta\eta} - u_{\eta z} - \frac{1}{2}u_{zz} + \frac{1}{2}u_{zz}$; $u_{yz} = u_{zz} - u_{zz} \Rightarrow$

$u_{zz} + u_{\eta\eta} + u_{zz} + 2u_{z\eta} - 2u_{zz} - 2u_{\eta z} - 2u_{z\eta} - 2u_{\eta\eta} + 4u_{\eta z} +$

$+ 2u_{z\eta} - 2u_{zz} - 2u_{\eta z} + 2u_{zz} + u_\eta - u_z + u_z = u_{zz} - u_{\eta\eta} +$

$+ u_{zz} + u_\eta = 0 \Rightarrow$ гиперболический тип.

Т I.2.

$u_{xx} + 2\alpha u_{xz} + u_{yy} + 4\beta u_{yz} + 4u_{zz} = 0 \Rightarrow A = \begin{pmatrix} 1 & 0 & \alpha \\ 0 & 1 & 2\beta \\ \alpha & 2\beta & 4 \end{pmatrix}$.

$\Delta_1 = 1 > 0$; $\Delta_2 = 1 > 0$; $\Delta_3 = 4 - 4\beta^2 - \alpha^2 \Rightarrow$

Параболический: $\Delta_3 = 0$, т.е. $\alpha^2 + 4\beta^2 = 4$;

Эллиптический: $\Delta_3 > 0$, т.е. $\alpha^2 + 4\beta^2 < 4$;

Гиперболический: $\Delta_3 < 0$, т.е. $\alpha^2 + 4\beta^2 > 4$.

№ 2.11(6).

$u_{xy} + 2xy u_y - 2xu = 0$. $D = \frac{1}{4} > 0 \Rightarrow$ тип гиперболический

во всей плоскости (x,y). $v := u_y \Rightarrow v_x + 2xyv - 2xu = 0 \Rightarrow$

$u = yv + \frac{v_x}{2x}$. $v_{xy} + 2xv + 2xyv_y - 2xv = 0 \Rightarrow v_{xy} + 2xyv_y = 0$

$w := v_y \Rightarrow w_x + 2xyw = 0 \Rightarrow \frac{dw}{w} = -2xy\,dx \Rightarrow w = f(y)\cdot e^{-x^2y} \Rightarrow$

$v = \int_0^y f(\xi) e^{-x^2\xi}\,d\xi + g(x) \Rightarrow v_x = -2x\int_0^y \xi f(\xi) e^{-x^2\xi}\,d\xi + g'(x) \Rightarrow$

$u = \frac{g'(x)}{2x} + yg(x) + \int_0^y (y-\xi) f(\xi) e^{-x^2\xi}\,d\xi$.

Т II.1.

а). $y^3 u_{xy} - y u_{yy} - 3y^5 u_x + (2+3y^3) u_y = 0$; $y > 0$; $x \in \mathbb{R}$;

$u|_{y=1} = 1 + 3x$; $u_y|_{y=1} = 3(4+3x)$; $x \in \mathbb{R}$.

Хар. ур.: $-y^3 dx dy - y(dx)^2 = 0 \Rightarrow$ 1). $x = C_1$; 2). $y^3 y' + y = 0 \Rightarrow y = 0$

или $y^2 y' = -1 \Rightarrow y^3 + 3x = C_2 \Rightarrow \begin{cases} \xi = x \\ \eta = y^3 + 3x \end{cases} \Rightarrow u_x = u_\xi + 3u_\eta$;

$u_y = 3y^2 u_\eta$; $u_{xy} = 3y^2 u_{\xi\eta} + 3y^2 u_{\eta\eta}$; $u_{yy} = 6y u_\eta + 9y^4 u_{\eta\eta} \Rightarrow$

$y^3(3y^2 u_{\xi\eta} + 3y^2 u_{\eta\eta}) - 6y^2 u_\eta - 9y^5 u_{\eta\eta} - 3y^5(u_\xi + 3u_\eta) + (2+3y^3) 3y^2 u_\eta = 0 \Rightarrow$

$y^5 u_{\xi\eta} - y^5 u_\xi = 0 \Rightarrow \frac{\partial u_\xi}{\partial \eta} = u_\xi \Rightarrow u_\xi = f(\xi) e^\eta \Rightarrow u = e^\eta \int f(\xi) d\xi +$

$+ g(\eta) = e^\eta f_1(\xi) + g(\eta) \Rightarrow u = e^{y^3+3x} f_1(x) + g(y^3+3x)$

Нач. усл.: $e^{3x+1} f_1(x) + g(3x+1) = 3x+1$; $u_y|_{y=1} = (3y^2 e^{y^3+3x} f_1(x) +$

$+ 3y^2 g'(y^3+3x))|_{y=1} = 3e^{1+3x} f_1(x) + 3g'(1+3x) = 3(1+3x)$

$h(x) := e^{3x+1} f_1(x) \Rightarrow \begin{cases} h(x) + g(3x+1) = 3x+1 \\ h(x) + g'(3x+1) = 1+3x \end{cases} \Rightarrow g'(3x+1) = 1 - \frac{h'(x)}{3} \Rightarrow$

$h(x) - \frac{h'(x)}{3} = 3x+3 \Rightarrow$ Одн.: $h_0(x) = Ce^{3x}$; Част.: $h_1(x) = ax+b \Rightarrow$

$ax + b - \frac{a}{3} = 3x+3 \Rightarrow \begin{cases} a = 3 \\ b = 4 \end{cases} \Rightarrow h(x) = Ce^{3x} + 3x + 4 \Rightarrow g(3x+1) = -Ce^{3x} - 3 \Rightarrow$

$g(z) = -3 - Ce^{z-1} \Rightarrow u = e^{y^3+3x} \frac{Ce^{3x}+3x+4}{e^{3x+1}} - 3 - Ce^{y^3+3x-1} = e^{y^3-1}(3x+4) - 3$.

б). $x^2 u_{xx} - 9y^2 u_{yy} + 3x u_x - 3y u_y = 0$; $x > 1$; $y > 1$;

$u|_{x=y} = y^{2/3}$, $u_x|_{x=y} = y^{-3} + y^{-1/3}$, $y > 1$. Хар. ур-я: $x^2 (dy)^2 - 9y^2 (dx)^2 = 0$

$(x dy - 3y dx)(x dy + 3y dx) = 0 \Rightarrow$ 1). $3\frac{dx}{x} = -\frac{dy}{y} \Rightarrow y = C_1 x^3$;

2). $y = C_2 x^{-3} \Rightarrow \begin{cases} \xi = y/x^3 \\ \eta = yx^3 \end{cases} \Rightarrow u_x = -\frac{3y}{x^4} u_\xi + 3yx^2 u_\eta$;

$u_y = \frac{u_\xi}{x^3} + x^3 u_\eta$; $u_{xx} = \frac{12y}{x^5} u_\xi + 6xy u_\eta + \frac{9y^2}{x^8} u_{\xi\xi} + 9y^2 x^4 u_{\eta\eta} - \frac{18y^2}{x^2} u_{\xi\eta}$;

$u_{yy} = \frac{u_{\xi\xi}}{x^6} + x^6 u_{\eta\eta} + 2u_{\xi\eta} \Rightarrow \frac{12y}{x^3} u_\xi + 6yx^3 u_\eta + \frac{9y^2}{x^6} u_{\xi\xi} + 9y^2 x^6 u_{\eta\eta} -$

$-18y^2 u_{\xi\eta} - 9y^2 \frac{u_{\xi\xi}}{x^6} - 9y^2 x^6 u_{\eta\eta} - 18y^2 u_{\xi\eta} - \frac{9y}{x^3} u_\xi + 9yx^3 u_\eta - 3y\frac{u_\xi}{x^3} -$

$-3yx^3 u_\eta = 0 \Rightarrow -3y^2 u_{\xi\eta} + yx^3 u_\eta = 0 \Rightarrow -3\xi\eta u_{\xi\eta} + \eta u_\eta = 0 \Rightarrow$

$3\frac{\partial u_z}{u_z}=\frac{\partial z}{z}\Rightarrow u_z=z^{1/3}f(\eta)\Rightarrow u=z^{1/3}f_1(\eta)+g(\frac{z}{\eta})=\frac{y^{1/3}}{x}f_1(yx^3)+$
$+g(\frac{y}{x^3})\Rightarrow u_x=-\frac{y^{1/2}}{x^2}f_1(yx^3)+3xy^{4/3}f_1'(yx^3)-3\frac{y}{x^4}g'(\frac{y}{x^3})$.
Нач. усл.: $u|_{x=y}=y^{-2/3}f_1(y^4)+g(y^{-2})=y^{2/3}$; $u_x|_{x=y}=-y^{-5/3}f_1(y^4)+$
$+3y^{7/3}f_1'(y^4)-3y^{-3}g'(y^{-2})=y^{-3}+y^{-1/3}\Rightarrow\frac{2}{3}y^{-1/3}=-\frac{2}{3}y^{-5/3}f_1(y^4)+$
$+4y^{7/3}f_1'(y^4)-2y^{-3}g'(y^{-2})\Rightarrow g'(y^{-2})=\frac{1}{2}y^3\left(-\frac{2}{3}y^{-1/3}-\frac{2}{3}y^{-5/3}f_1(y^4)+4y^{7/3}f_1'(y^4)\right)$
$-y^{-5/3}f_1(y^4)+3y^{7/3}f_1'(y^4)+y^{-1/3}+y^{-1/3}+y^{-5/3}f_1(y^4)-6y^{7/3}f_1'(y^4)=y^{-3}+y^{-1/3}\Rightarrow$
$-3y^{7/3}f_1'(y^4)=y^{-3}\Rightarrow f_1'(y^4)=-\frac{1}{3}y^{-16/3}\Rightarrow f_1'(z)=-\frac{1}{3}z^{-4/3}\Rightarrow$
$f_1(z)=z^{-1/3}+C\Rightarrow g(y^{-2})=y^{2/3}-y^{-2}-Cy^{-2/3}\Rightarrow g(t)=t^{-1/3}-t-Ct^{1/3}\Rightarrow$
$u(x,y)=\frac{y^{1/3}}{x}(x^{-1}y^{-1/3}+C)+\frac{x}{y^{1/3}}-\frac{y}{x^3}-C\frac{y^{1/3}}{x}=\frac{1}{x^2}+\frac{x}{y^{1/3}}-\frac{y}{x^3}$.

б). $x^2u_{xx}-xyu_{xy}-2y^2u_{yy}+xu_x-2yu_y=9xy^2$, $x>0$, $y>0$;
$u|_{x=1}=3e^y$; $u_x|_{x=1}=-y^2$. Ур. хар-к: $x^2(dy)^2+xy\,dx\,dy-2y^2(dx)^2=0$
$x^2\left(\frac{dy}{dx}\right)^2+xy\left(\frac{dy}{dx}\right)-2y^2=0\Rightarrow\frac{dy}{dx}=\frac{-xy\pm3xy}{2x^2}\Rightarrow\begin{bmatrix}y=C_1x\\y=C_2x^{-2}\end{bmatrix}\Rightarrow$
$\begin{matrix}\xi=y/x\\ \eta=x^2y\end{matrix}\Rightarrow u_x=-\frac{y}{x^2}u_\xi+2xyu_\eta$; $u_y=\frac{u_\xi}{x}+x^2u_\eta$; $u_{xx}=\frac{2y}{x^3}u_\xi+$
$+2yu_\eta+\frac{y^2}{x^4}u_{\xi\xi}+4x^2y^2u_{\eta\eta}-4\frac{y^2}{x}u_{\xi\eta}$; $u_{yy}=\frac{u_{\xi\xi}}{x^2}+x^4u_{\eta\eta}+2xu_{\xi\eta}$;
$u_{xy}=-\frac{u_\xi}{x^2}+2xu_\eta-\frac{y}{x^3}u_{\xi\xi}+yu_{\xi\eta}+2x^3yu_{\eta\eta}\Rightarrow x^2\left(\frac{2y}{x^3}u_\xi+2yu_\eta+\right.$
$\left.+\frac{y^2}{x^4}u_{\xi\xi}+4x^2y^2u_{\eta\eta}-4\frac{y^2}{x}u_{\xi\eta}\right)-xy\left(-\frac{u_\xi}{x^2}+2xu_\eta+yu_{\xi\eta}-\frac{y}{x^3}u_{\xi\xi}+\right.$
$\left.+2x^3yu_{\eta\eta}\right)-2y^2\left(\frac{u_{\xi\xi}}{x^2}+x^4u_{\eta\eta}+2xu_{\xi\eta}\right)+x\left(-\frac{y}{x^2}u_\xi+2xyu_\eta\right)-2y\left(\frac{u_\xi}{x}+\right.$
$\left.+x^2u_\eta\right)=9xy^2\Rightarrow-9xy^2u_{\xi\eta}=9xy^2\Rightarrow u_{\xi\eta}=-1\Rightarrow u=-\xi\eta+f(\xi)+g(\eta)=$
$=-xy^2+f(y/x)+g(x^2y)$. $u_x=-y^2-\frac{y}{x^2}f'(\frac{y}{x})+2xyg'(x^2y)$.
Нач. усл.: $u|_{x=1}=-y^2+f(y)+g(y)=3e^y$; $u_x|_{x=1}=-y^2-yf'(y)+2yg'(y)=$

$=-y^2 \Rightarrow f'(y)=2g'(y)$; $f'(y)+g'(y)=3e^y+2y \Rightarrow 3g'(y)=3e^y+2y \Rightarrow$
$3g(y)=3e^y+y^2+C$; $f(y)=2e^y+\frac{2y^2}{3}-\frac{C}{3} \Rightarrow u=-xy^2+2e^{y/x}+\frac{2}{3}\left(\frac{y}{x}\right)^2+e^{x^3y}+x^4\frac{y^2}{3}$.

2). $yu_{xx}+(x-y)u_{xy}-xu_{yy}-u_x+u_y=0$; $u|_{y=0}=2x^2$; $u_y|_{y=0}=2x$; $1<x<4$.

Ур. хар-к: $y(dy)^2+(y-x)\,dx\,dy-x(dx)^2=0 \Rightarrow y\left(\frac{dy}{dx}\right)^2+(y-x)\frac{dy}{dx}-x=0 \Rightarrow$
$\frac{dy}{dx}=\frac{x-y\pm(x+y)}{2y} \Rightarrow \begin{bmatrix} y^2=x^2+c_1 \\ y=-x+c_2 \end{bmatrix} \Rightarrow \begin{cases} \xi=y^2-x^2 \\ \eta=y+x \end{cases} \Rightarrow u_x=-2xu_\xi+u_\eta$;

$u_y=2yu_\xi+u_\eta$; $u_{xx}=-2u_\xi+4x^2u_{\xi\xi}+u_{\eta\eta}-4xu_{\xi\eta}$; $u_{yy}=2u_\xi+4y^2u_{\xi\xi}+$
$+u_{\eta\eta}+4yu_{\xi\eta}$; $u_{xy}=-4xyu_{\xi\xi}+u_{\eta\eta}-2xu_{\xi\eta}+2yu_{\xi\eta} \Rightarrow y(-2u_\xi+4x^2u_{\xi\xi}+$
$+u_{\eta\eta}-4xu_{\xi\eta})+(x-y)(-4xyu_{\xi\xi}+u_{\eta\eta}-2xu_{\xi\eta}+2yu_{\xi\eta})-x(2u_\xi+4y^2u_{\xi\xi}+$
$+u_{\eta\eta}+4yu_{\xi\eta})+2xu_\xi-u_\eta+2yu_\xi+u_\eta=0 \Rightarrow (x+y)^2u_{\xi\eta}=0 \Rightarrow$
$u_{\xi\eta}=0 \Rightarrow u=f(\xi)+g(\eta)=f(y^2-x^2)+g(x+y) \Rightarrow u_y=2yf'(y^2-x^2)+$
$+g'(x+y)$. Нач. усл.: $u|_{y=0}=f(-x^2)+g(x)=2x^2$; $u_y|_{y=0}=g'(x)=2x \Rightarrow$
$g(x)=x^2+C$; $f(-x^2)=x^2-C \Rightarrow u=x^2-y^2-C+(x+y)^2+C=2x^2+2xy$.

g). $y^4u_{yy}+y^2u_{xy}-2u_{xx}+2y^3u_y=0$; $u|_{y=1}=x^2+5$, $u_y|_{y=1}=2x-6$, $1<x<2$.

Ур. хар-к: $y^4(dx)^2-y^2dxdy-2(dy)^2=0 \Rightarrow \begin{bmatrix} \frac{dx}{dy}=\frac{2}{y^2} \\ \frac{dx}{dy}=-\frac{1}{y^2} \end{bmatrix} \Rightarrow \begin{bmatrix} x=-\frac{2}{y}+c_1 \\ x=\frac{1}{y}+c_2 \end{bmatrix} \Rightarrow$
$\begin{cases} \xi=x+\frac{2}{y} \\ \eta=x-\frac{1}{y} \end{cases} \Rightarrow u_x=u_\xi+u_\eta$; $u_y=-\frac{2}{y^2}u_\xi+\frac{1}{y^2}u_\eta$; $u_{xx}=u_{\xi\xi}+u_{\eta\eta}+2u_{\xi\eta}$;
$u_{yy}=\frac{4}{y^3}u_\xi-\frac{2}{y^3}u_\eta+\frac{4}{y^4}u_{\xi\xi}+\frac{1}{y^4}u_{\eta\eta}-\frac{4}{y^4}u_{\xi\eta}$; $u_{xy}=-\frac{2}{y^2}u_{\xi\xi}+\frac{1}{y^2}u_{\eta\eta}-\frac{1}{y^2}u_{\xi\eta} \Rightarrow$
$y^4\left(\frac{4}{y^3}u_\xi-\frac{2}{y^3}u_\eta+\frac{4}{y^4}u_{\xi\xi}+\frac{1}{y^4}u_{\eta\eta}-\frac{4}{y^4}u_{\xi\eta}\right)+y^2\left(-\frac{2}{y^2}u_{\xi\xi}+\frac{1}{y^2}u_{\eta\eta}-\frac{1}{y^2}u_{\xi\eta}\right)-2(u_{\xi\xi}+$
$+u_{\eta\eta}+2u_{\xi\eta})+2y^3\left(-\frac{2}{y^2}u_\xi+\frac{1}{y^2}u_\eta\right)=0 \Rightarrow -9u_{\xi\eta}=0 \Rightarrow u=f(\xi)+g(\eta)=$
$=f\left(x+\frac{2}{y}\right)+g\left(x-\frac{1}{y}\right)$. $u_y=-\frac{2}{y^2}f'\left(x+\frac{2}{y}\right)+\frac{1}{y^2}g'\left(x-\frac{1}{y}\right)$. Нач. усл.:
$u|_{y=1}=f(x+2)+g(x-1)=x^2+5$; $u_y|_{y=1}=-2f'(x+2)+g'(x-1)=2x-6$.
$f'(x+2)+g'(x-1)=2x \Rightarrow -4x+3g'(x-1)=2x-6 \Rightarrow g'(x-1)=2x-2 \Rightarrow$

$g(x-1)=x^2-2x+C$; $f(x+2)=2x+5-C \Rightarrow u=2\left(x+\frac{2}{y}-2\right)+5+\left(x-\frac{1}{y}+1\right)^2-$
$-2\left(x-\frac{1}{y}+1\right)=x^2+2x-\frac{2x}{y}+\frac{1}{y^2}+\frac{4}{y}$.

ТП.2.

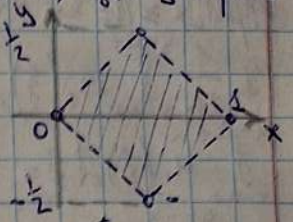
$x^2u_{xx}-4y^2u_{yy}+xu_x-4yu_y=0$; $\frac{1}{x^2}<y<x^2$, $x>0$; $u|_{y=\frac{1}{x^2}}=1+2x^4$;
$u|_{y=x^2}=2+x^4$. Ур. хар-к: $x^2(dy)^2-4y^2(dx)^2=0 \Rightarrow$
$(xdy-2ydx)(xdy+2ydx)=0 \Rightarrow$ $\begin{bmatrix} y=C_1x^2 \\ y=C_2x^{-2} \end{bmatrix} \Rightarrow \begin{cases} \xi=y/x^2 \\ \eta=x^2y \end{cases} \Rightarrow$
$u_x=-\frac{2y}{x^3}u_\xi+2xyu_\eta$; $u_y=\frac{u_\xi}{x^2}+x^2u_\eta$; $u_{xx}=\frac{6y}{x^4}u_\xi+2yu_\eta+$
$+\frac{4y^2}{x^6}u_{\xi\xi}+4y^2x^2u_{\eta\eta}-\frac{8y^2}{x^2}u_{\xi\eta}$; $u_{yy}=\frac{u_{\xi\xi}}{x^4}+x^4u_{\eta\eta}+2u_{\xi\eta} \Rightarrow$
$x^2\left(\frac{6y}{x^4}u_\xi+2yu_\eta+\frac{4y^2}{x^6}u_{\xi\xi}+4y^2x^2u_{\eta\eta}-\frac{8y^2}{x^2}u_{\xi\eta}\right)-4y^2\left(\frac{u_{\xi\xi}}{x^4}+x^4u_{\eta\eta}+\right.$
$\left.+2u_{\xi\eta}\right)+x\left(-\frac{2y}{x^3}u_\xi+2xyu_\eta\right)-4y\left(\frac{u_\xi}{x^2}+x^2u_\eta\right)=0 \Rightarrow y^2u_{\xi\eta}=0 \Rightarrow$
$u=f(\xi)+g(\eta)=f\left(\frac{y}{x^2}\right)+g(x^2y) \Rightarrow$ Нач. усл.: $f\left(\frac{1}{x^4}\right)+g(1)=$
$1+2x^4$; $f(1)+g(x^4)=2+x^4 \Rightarrow f(1)+g(1)=3$. $C:=g(1) \Rightarrow$
$g(t)=C+t-1$; $f(s)=\frac{2}{s}+1-C \Rightarrow u(x,y)=\frac{2x^2}{y}+x^2y$.

ТП.3.

$u_{yy}-u_{xx}=0$, $u|_{y=0}=u_0(x)$, $u_y|_{y=0}=u_1(x)$, $0<x<1$, $u_0(x)\in C^2(0;1)$,
$u_1(x)\in C^1(0;1)$, решение $\exists!$. Ур. хар-к: $(dy-dx)(dy+dx)=0 \Rightarrow$
$\begin{bmatrix} y=x+C_1 \\ y=-x+C_2 \end{bmatrix} \Rightarrow \begin{cases} \xi=y-x \\ \eta=y+x \end{cases} \Rightarrow u_x=-u_\xi+u_\eta$, $u_{xx}=u_{\xi\xi}-2u_{\xi\eta}+u_{\eta\eta}$, $u_y=u_\xi+u_\eta$,
$u_{yy}=u_{\xi\xi}+2u_{\xi\eta}+u_{\eta\eta} \Rightarrow u_{\xi\eta}=0 \Rightarrow u=f(y-x)+g(y+x) \Rightarrow$
$u_0(x)=f(-x)+g(x)$; $u_1(x)=g'(x)+f'(-x) \Rightarrow g'(x)=\frac{u_1(x)+u_0(x)}{2} \Rightarrow$
$g(x)=\frac{u_0}{2}+\int_{x_0}^{x}\frac{u_1(s)}{2}ds+C \Rightarrow f(-x)=\frac{u_0(x)}{2}-\int_{x_0}^{x}\frac{u_1(s)}{2}ds-C \Rightarrow$
$u(x,y)=\frac{u_0(x-y)}{2}-\int_{x_0}^{x-y}\frac{u_1(s)}{2}ds+\frac{u_0(x+y)}{2}+\int_{x_0}^{x+y}\frac{u_1(s)}{2}ds=\frac{u_0(x+y)+u_0(x-y)}{2}+$
$+\frac{1}{2}\int_{x-y}^{x+y}u_1(s)ds$ – ф-ла Даламбера

Решение определено однозначно в точках $(x;y)$, для которых область зависимости целиком лежит внутри $(0;1)$, т.е. $u_0(x+y)$ опр. $\Rightarrow 0<x+y<1$; $u_0(x-y)$ опр. $\Rightarrow$ $0<x-y<1$; $\int_{x-y}^{x+y} u_1(s)\,ds$ опр. $\Rightarrow [x-y;x+y]\subset(0;1)\Rightarrow$

$\begin{cases}0<x+y<1\\0<x-y<1\end{cases}$ т.е. искомая область ограничена квадратом с вершинами $(0;0)$, $(\frac12;\frac12)$, $(1;0)$, $(\frac12;-\frac12)$.



ТIII.1.

$\begin{cases}4u_{tt}=u_{xx}+4t^2\cos 2x,\ (t,x)\in\mathbb{R}^2\\u|_{t=0}=e^x,\ u_t|_{t=0}=x^2,\ x\in\mathbb{R}\end{cases}\Rightarrow u(x,t)=\frac{e^{x-\frac{t}{2}}+e^{x+\frac{t}{2}}}{2}+\frac{1}{2\cdot\frac12}\int_{x-\frac{t}{2}}^{x+\frac{t}{2}}s^2\,ds+$

$+\frac{1}{2\cdot\frac12}\int_0^t\int_{x-\frac12(t-\tau)}^{x+\frac12(t-\tau)}\tau^2\cos 2s\,ds\,d\tau=e^x\operatorname{ch}\frac{t}{2}+\frac13\left(\left(x+\frac{t}{2}\right)^3-\left(x-\frac{t}{2}\right)^3\right)+\int_0^t\frac{\tau^2}{2}\left(\sin(2x+t-\tau)-\right.$

$\left.-\sin(2x-t+\tau)\right)d\tau=e^x\operatorname{ch}\frac{t}{2}+x^2t+\frac{t^3}{12}+\cos 2x\left(t^2-2+\cos 2t\right)$.

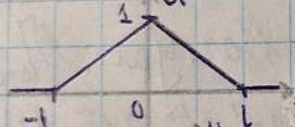
ТIII.2.

$u_{tt}=a^2u_{xx}$, $x\in\mathbb{R}$, $t>0$. а). $u|_{t=0}=\varphi(x)$, $u_t|_{t=0}=0$. б). $u|_{t=0}=0$, $u_t|_{t=0}=\psi(x)$.

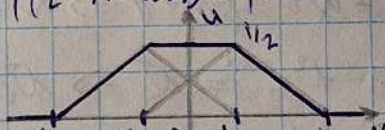
$\varphi(x)=\left(1-\frac{|x|}{l}\right)\theta(l-|x|)$, $\psi(x)=\theta(l-|x|)$, $l=\text{const}>0$, $\theta(x)=\begin{cases}1, & x\ge0\\0, & x<0\end{cases}$.

а). $u(x,t)=\frac{\varphi(x-at)+\varphi(x+at)}{2}=\frac{1}{2l}\left((l-|x-at|)\theta(l-|x-at|)+(l-|x+at|)\theta(l-|x+at|)\right)$.

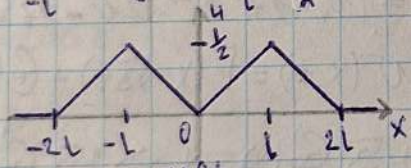
$t=0$:



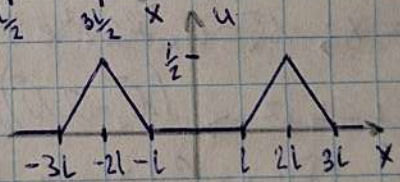
$t=\frac{l}{2a}$:



$t=\frac{l}{a}$:

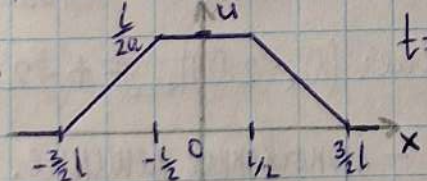


$t=\frac{2l}{a}$:

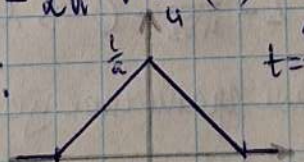


б). $u(x,t)=\frac{1}{2a}\int_{x-at}^{x+at}\theta(l-|s|)\,ds=\frac{1}{2a}\cdot\max(0,\min(x+at,l)-\max(x-at,-l))$

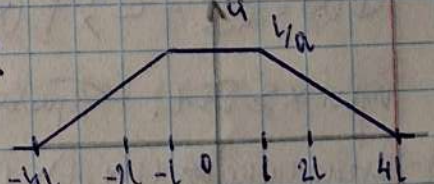
$t=\frac{l}{2a}$:



$t=\frac{l}{a}$:



$t=\frac{2l}{a}$:



✓ 21.2.

$u_{tt}=a^2u_{xx}$, $t>0$, $x>0$; $u|_{t=0}=u_0(x)$, $u_t|_{t=0}=u_1(x)$, $u|_{x=0}=0$; $u_0\in C^2(x\ge0)$,

$u_1\in C^1(x\ge0)$, $u_0(0)=u_0''(0)=u_1(0)=0$. Продолжим нечётно на $x\le0$:

$\tilde u_0(x)=\begin{cases}u_0(x), & x\ge0\\-u_0(-x), & x<0\end{cases}$, $\tilde u_1(x)=\begin{cases}u_1(x), & x\ge0\\-u_1(-x), & x<0\end{cases}$. Из усл. $\Rightarrow \tilde u_0\in C^2(\mathbb{R})$, $\tilde u_1\in C^1(\mathbb{R})$.

по ф-ле Даламбера: $\tilde{u}(x,t)=\frac{1}{2}(\tilde{u}_0(x+at)+\tilde{u}_0(x-at))+\frac{1}{2a}\int_{x-at}^{x+at}\tilde{u}_1(\xi)d\xi$.

при $\begin{cases}x>0\\ x\geq at\end{cases}$ $\hookrightarrow x+at\geq 0,\ x-at\geq 0 \Rightarrow u(x,t)=\frac{1}{2}(u_0(x+at)+u_0(x-at))+$

$+\frac{1}{2a}\int_{x-at}^{x+at}u_1(\xi)d\xi$. при $\begin{cases}x>0\\ x<at\end{cases}$ $\hookrightarrow x-at<0 \Rightarrow \tilde{u}_0(x-at)=-u_0(at-x)$,

$\int_{x-at}^{x+at}\tilde{u}_1(\xi)d\xi=\int_{x-at}^{0}\tilde{u}_1(\xi)d\xi+\int_{0}^{x+at}\tilde{u}_1(\xi)d\xi=-\int_{x-at}^{0}u_1(-\xi)d\xi+\int_{0}^{x+at}u_1(\xi)d\xi=$

$=-\int_{at-x}^{0}u_1(\xi)d\xi+\int_{0}^{x+at}u_1(\xi)d\xi=\int_{x-at}^{x+at}u_1(\xi)d\xi$ (нижний предел $x-at$, поскольку мы подразумеваем $u_1(x)$ некоё продолжение на $x\leq 0$).

Значит, решение $u(x,t)=\begin{cases}\frac{1}{2}(u_0(x+at)+u_0(x-at))+\frac{1}{2a}\int_{x-at}^{x+at}u_1(\xi)d\xi, & x\geq at\\ \frac{1}{2}(u_0(x+at)-u_0(x-at))+\frac{1}{2a}\int_{x-at}^{x+at}u_1(\xi)d\xi, & x<at\end{cases}$

единственно на ограничении $x>0$, поскольку оно единственно на $x\in\mathbb{R}$.

№21.19.

$u_{tt}=3u_{xx}+2(1-6t^2)e^{-2x}$, $t>0$, $x>0$; $u|_{t=0}=1$; $u_t|_{t=0}=x$, $(u_x-2u)|_{x=0}=-2+t-4t^2$.

$u_2^{(x,t)}=a(t)e^{-2x} \Rightarrow a''(t)e^{-2x}=12a(t)e^{-2x}+2(1-6t^2)e^{-2x} \Rightarrow a''(t)=12a(t)-12t^2+2 \Rightarrow$

$a(t)=C_1e^{2\sqrt{3}t}+C_2e^{-2\sqrt{3}t}+\alpha t^2+\beta t+\gamma \Rightarrow \alpha=1 \Rightarrow$ возьмём $a(t)=t^2 \Rightarrow$

$u_2=t^2e^{-2x}$; $u_0=u-t^2e^{-2x}=:v \Rightarrow v_{tt}=3v_{xx}$. Нач. усл.: $v|_{t=0}=1$;

$v_t|_{t=0}=x$; $v_x=u_x+2t^2e^{-2x} \Rightarrow (u_x-2u)_{x=0}=(v_x-2t^2e^{-2x}-2v-2t^2e^{-2x})|_{x=0}=$

$=-2+t-4t^2 \Rightarrow (v_x-2v)|_{x=0}=t-2$. $v(x,t)=F(x-\sqrt{3}t)+G(x+\sqrt{3}t) \Rightarrow$

$v|_{t=0}=F(x)+G(x)=1$; $v_t|_{t=0}=-\sqrt{3}F'(x)+\sqrt{3}G'(x)=x \Rightarrow 2\sqrt{3}G'(x)=x \Rightarrow$

$G(x)=\frac{x^2}{4\sqrt{3}}+C$; $F(x)=1-\frac{x^2}{4\sqrt{3}}-C \Rightarrow v(x,t)=1-\frac{(x-\sqrt{3}t)^2}{4\sqrt{3}}+\frac{(x+\sqrt{3}t)^2}{4\sqrt{3}}=1+xt$

при $x\geq\sqrt{3}t$. Проверим граничное условие: $(v_x-2v)|_{x=0}=t-2$ – выполнено $\Rightarrow u(x,t)=v+t^2e^{-2x}=1+xt+t^2e^{-2x}$ – единственное решение.

Т III.3.

б). $u_{tt}=u_{xx}+xe^t$, $x>0$, $t>0$; $u|_{t=0}=1+x$, $u_t|_{t=0}=4-5x$, $x\geq 0$; $(2u+u_x)|_{x=0}=$

$=(1+t)e^t+2+t-3t^2$, $t\geq 0$. $u_2(x,t)=a(x)e^t \Rightarrow u_{tt}=a(x)e^t \Rightarrow a''(x)e^t+$

$+xe^t=a(x)e^t \Rightarrow u_2(x,t)=xe^t$. $u_0=u-xe^t=:v \Rightarrow v_{tt}=v_{xx}$.

Нач. усл.: $v|_{t=0}=u|_{t=0}-x=1$; $v_t|_{t=0}=4-6x$; $(2v+2xe^t+v_x+e^t)|_{x=0}=(1+t)e^t+$

$+2+t-3t^2 \Rightarrow (2v+v_x)|_{x=0} = te^t+2+t-3t^2$. $v(x,t) = F(x-t)+G(x+t) \Rightarrow$
$F(x)+G(x)=1$; $-F'(x)+G'(x)=4-6x \Rightarrow G'(x)=2-3x \Rightarrow G(x)=2x-\frac{3}{2}x^2+C$;
$F(x)=1-2x+\frac{3}{2}x^2-C \Rightarrow$ при $x \geq t \hookrightarrow v(x,t)=1-2(x-t)+\frac{3}{2}(x-t)^2+2(x+t)-$
$-\frac{3}{2}(x+t)^2 = 1+4t-6xt$. Граничное условие: $(2v+v_x)|_{x=0} = 2+2t$ — не выполнено $\Rightarrow$
Рассматриваем случай $x<t$: $v(0,t)=F(-t)+G(t) \Rightarrow (2v+v_x)|_{x=0} = 2(F(-t)+$
$+G(t))+(F'(-t)+G'(t)) = te^t+2+t-3t^2 \Rightarrow 2F(-t)+F'(-t) = te^t-2C \Rightarrow$
$2F(z)+F'(z) = -ze^z-2C \Rightarrow F(z) = (1-z)e^z-C+C_2e^{-2z} \Rightarrow v(x,t) = (1-x+t)e^{t-x}-$
$-C+C_2e^{2(t-x)}+2(x+t)-\frac{3}{2}(x+t)^2+C$. $F(0)=1-C+C_2=1-C \Rightarrow C_2=0$
(по непрерывности) $\Rightarrow v(x,t) = \begin{cases} 1+4t-6xt, & x \geq t \\ (1-x+t)e^{t-x}+2(x+t)-\frac{3}{2}(x+t)^2, & x<t \end{cases}$; $u(x,t) = v(x,t)+xe^t$.

1) $4u_{tt} = u_{xx}-3\sin(x+t)$, $x>0$, $t>0$; $u|_{t=0} = \sin x+\sin 2x$, $u_t|_{t=0} = 2+\cos x$, $x \geq 0$;
$(u_x-2u)|_{x=0} = 2-4t-2\sin t+\cos t$, $t \geq 0$. $u_2(x,t) = \alpha\sin(x+t)+$
$+\beta\cos(x+t) \Rightarrow \alpha=1, \beta=0 \Rightarrow u_2(x,t) = \sin(x+t)$. $u_0(x,t) = u(x,t)-$
$-\sin(x+t) =: v \Rightarrow 4v_{tt} = v_{xx}$. Нач. усл.: $v|_{t=0} = \sin 2x$;
$v_t|_{t=0} = (u_t-\cos(x+t))|_{t=0} = 2$; $(u_x-2u)|_{x=0} = (v_x+\cos(x+t)-2v-$
$-2\sin(x+t))|_{x=0} \Rightarrow (v_x-2v)|_{x=0} = 2-4t$. $v(x,t) = F(x-\frac{t}{2})+$
$+G(x+\frac{t}{2})$. $v|_{t=0} = F(x)+G(x) = \sin 2x$; $v_t|_{t=0} = -\frac{1}{2}F'(x)+\frac{1}{2}G'(x) = 2 \Rightarrow$
$G'(x) = \cos 2x+2 \Rightarrow G(x) = \frac{1}{2}\sin 2x+2x+C \Rightarrow F(x) = \frac{1}{2}\sin 2x-2x-C \Rightarrow$
при $x \geq \frac{t}{2} \hookrightarrow v(x,t) = \frac{1}{2}(\sin(2x+t)+\sin(2x-t))+2x+t-2x+t = 2t+$
$+\sin 2x\cos t$. Граничное условие: $(v_x-2v)|_{x=0} = 2\cos t-4t \neq 2-4t$ —
не выполнено $\Rightarrow$ рассматриваем $x<\frac{t}{2}$: $(v_x-2v)|_{x=0} = F'(-\frac{t}{2})+G'(\frac{t}{2})-$
$-2F(-\frac{t}{2})-2G(\frac{t}{2}) = F'(-\frac{t}{2})-2F(\frac{t}{2})-2C-\sin t-2t+2t+\cos t = 2-4t$.
$F'(z)-2F(z) = -\cos 2z-\sin 2z+4z+2C \Rightarrow F(z) = C_1e^{2z}+\frac{1}{2}\cos 2z-2z-1-C$.
по непр. $F(z)$: $F(0) = C_1+\frac{1}{2}-1-C = -C \Rightarrow C_1 = \frac{1}{2} \Rightarrow$

$u(x,t) = \begin{cases} 2t+\sin 2x\cos t+\sin(x+t), & x \geq t/2 \\ \frac{1}{2}(\sin(2x+t)+\cos(2x-t))+\sin(x+t)+2t-1+\frac{1}{2}e^{2x-t}, & x<t/2 \end{cases}$.

4). $u_{tt}+u_{xt}-2u_{xx} = 0$, $x>0$, $t>0$; $u|_{t=0} = \operatorname{sh}x+\operatorname{arctg}x$, $u_t|_{t=0} = \operatorname{ch}x-\frac{2}{1+x^2}$,
$x \geq 0$; $(u_x)|_{x=0} = \operatorname{ch}t+1$, $t \geq 0$.

Ур. хар-к: $(dx)^2 - dx\,dt - 2(dt)^2 = 0 \Leftrightarrow (x'-2)(x'+1)=0 \Rightarrow \begin{bmatrix} x = 2t + C_1 \\ x = -t + C_2 \end{bmatrix} \Rightarrow$

$\begin{cases} \xi = x - 2t \\ \eta = x + t \end{cases}$ $\Rightarrow$ решение представимо в виде $u(x,t) = F(x-2t) + G(x+t) \Rightarrow$

$F(x) + G(x) = \operatorname{sh} x + \operatorname{arctg} x$; $-2F'(x) + G'(x) = \operatorname{ch} x - \frac{2}{1+x^2} \Rightarrow F'(x) + G'(x) = \operatorname{ch} x + \frac{1}{1+x^2} \Rightarrow G(x) = \operatorname{sh} x + C$, $F(x) = \operatorname{arctg} x - C \Rightarrow u(x,t) = \operatorname{arctg}(x-2t) + \operatorname{sh}(x+t)$

при $x > 2t$. $u_x = F'(x-2t) + G'(x+t) \Rightarrow u_x|_{x=0} = F'(-2t) + G'(t) = \operatorname{ch} t + 1 \Rightarrow$

$F(z) = z + C_3$. $F(z)$ непр. в $0 \Rightarrow \operatorname{arctg} x \xrightarrow[x\to 0+]{} x \xrightarrow[x\to -0]{} + C_3 \Rightarrow C_3 = 0 \Rightarrow$

при $x \le 2t \hookrightarrow u(x,t) = x - 2t + \operatorname{sh}(x+t)$.

№ 12.45 (5).

$u_{tt} = 2\Delta u$; $u|_{t=0} = 2x^2 - y^2$; $u_t|_{t=0} = 2x^2 + y^2$. 1). $u_{tt} = 2u_{xx}$, $u|_{t=0} = 2x^2$, $u_t|_{t=0} = 2x^2 \Rightarrow u_1(x,t) = (x-\sqrt{2}t)^2 + (x+\sqrt{2}t)^2 + \frac{1}{\sqrt{2}} \int_{x-\sqrt{2}t}^{x+\sqrt{2}t} \xi^2\,d\xi = 2x^2 + 4t^2 + \frac{1}{3\sqrt{2}}\left((x+\sqrt{2}t)^3 - (x-\sqrt{2}t)^3\right) = 2x^2 + 4t^2 + 2x^2 t + \frac{4}{3}t^3$. 2). $u_{tt} = 2u_{yy}$, $u|_{t=0} = -y^2$, $u_t|_{t=0} = y^2 \Rightarrow u_2(y,t) = \frac{1}{2}\left(-(y-\sqrt{2}t)^2 - (y+\sqrt{2}t)^2\right) + \frac{1}{2\sqrt{2}} \int_{y-\sqrt{2}t}^{y+\sqrt{2}t} \xi^2\,d\xi =$

$= -y^2 - 2t^2 + \frac{1}{6\sqrt{2}}\left((y+\sqrt{2}t)^3 - (y-\sqrt{2}t)^3\right) = -y^2 - 2t^2 + y^2 t + \frac{2t^3}{3} \Rightarrow$

$u(x,y,t) = u_1(x,t) + u_2(y,t) = 2x^2 - y^2 + (2x^2 + y^2)t + 2t^2 + 2t^3$.

№ 12.44 (7).

$u_{tt} = a^2 \Delta u + \cos x \sin y\, e^z$; $u|_{t=0} = x^2 e^{y+z}$, $u_t|_{t=0} = \sin x\, e^{y+z}$. Частное решение:

$u_2 = A \cos x \sin y\, e^z \Rightarrow 0 = a^2(-2A \cos x \sin y\, e^z + A \cos x \sin y\, e^z) + \cos x \sin y\, e^z \Rightarrow A = \frac{1}{a^2}$.

$u_2 = \frac{1}{a^2} \cos x \sin y\, e^z \Rightarrow v := u - u_2 \Rightarrow v|_{t=0} = x^2 e^{y+z} - \frac{1}{a^2} \cos x \sin y\, e^z$, $v_t|_{t=0} = \sin x\, e^{y+z}$; $v_{tt} = a^2 \Delta v$. 1). $v|_{t=0} = x^2 e^{y+z}$, $v_t|_{t=0} = 0$. Ищем решение в виде $v_1 = F(x,t) e^{y+z} \Rightarrow F_{tt} = a^2 F_{xx} + 2a^2 F$; $F|_{t=0} = x^2$; $F_t|_{t=0} = 0 \Rightarrow$ ищем в виде $F(x,t) = A(t)x^2 + B(t)x + C(t) \Rightarrow A''(t)x^2 + B''(t)x + C''(t) = 2a^2 A(t) + 2a^2(A(t)x^2 + B(t)x + C(t)) \Rightarrow A'' = 2a^2 A \Rightarrow A(t) = C_1 e^{\sqrt{2}at} + C_2 e^{-\sqrt{2}at}$; $M.y.: C_1 + C_2 = 1$, $C_1 - C_2 = 0 \Rightarrow A(t) = \operatorname{ch}(\sqrt{2}at)$. $B'' = 2a^2 B$, $B(0) = B'(0) = 0 \Rightarrow B = 0$;

$C'' = 2a^2(A+C) \Rightarrow C'' - 2a^2 C = 2a^2 \operatorname{ch}(\sqrt{2}at)$, $C(0) = C'(0) = 0$. $\Rightarrow$ ищем $C(t) = E + \operatorname{ch}(\sqrt{2}at) + Ft\operatorname{sh}(\sqrt{2}at) \Rightarrow 2\sqrt{2}aE\operatorname{sh}(\sqrt{2}at) + 2a^2Et\operatorname{ch}(\sqrt{2}at) + 2\sqrt{2}aF\operatorname{ch}(\sqrt{2}at) + 2a^2Ft\operatorname{sh}(\sqrt{2}at) - 2a^2t(E\operatorname{ch}(\sqrt{2}at) + F\operatorname{sh}(\sqrt{2}at)) = 2a^2\operatorname{ch}(\sqrt{2}at) \Rightarrow$ $E = 0$, $F = \frac{a}{\sqrt{2}} \Rightarrow C(t) = \frac{a}{\sqrt{2}}t\operatorname{sh}(\sqrt{2}at) \Rightarrow v_1 = e^{y+z}\left(x^2\operatorname{ch}(\sqrt{2}at) + \frac{at}{\sqrt{2}}\operatorname{sh}(\sqrt{2}at)\right)$.

2) $v_{t=0} = -\frac{1}{a^2}\cos x \sin y e^z$, $v_t|_{t=0} = 0$. $\Rightarrow$ Ищем $v_2 = S(t)\cos x \sin y e^z \Rightarrow$ $S''(t) = -a^2 S(t)$; $S(0) = -\frac{1}{a^2}$; $S'(0) = 0 \Rightarrow S(t) = -\frac{1}{a^2}\cos at \Rightarrow v_3 = -\frac{\cos at}{a^2}\cos x \sin y e^z$.

3) $v_{t=0} = 0$, $v_t|_{t=0} = \sin x e^{y+z} \Rightarrow$ Ищем $v_3 = G(t)\sin x e^{y+z} \Rightarrow G''(t)\sin x = -G(t)\sin x + 2a^2 G(t)\sin x \Rightarrow G''(t) - a^2 G(t) = 0 \Rightarrow G(t) = C_1 e^{at} + C_2 e^{-at}$. $G(0) = 0$, $G'(0) = 1 \Rightarrow C_1 = -C_2$; $a(C_1 - C_2) = 1 \Rightarrow G(t) = \frac{1}{a}\operatorname{sh} at \Rightarrow v_3 = \frac{1}{a}\operatorname{sh} at \sin x e^{y+z}$. Таким образом, $u(x,y,z,t) = v_1 + v_2 + v_3 + u_2 = \frac{1}{a^2}(1 - \cos at)\cos x \sin y e^z + e^{y+z}\left(x^2\operatorname{ch}(\sqrt{2}at) + \frac{at}{\sqrt{2}}\operatorname{sh}(\sqrt{2}at) + \frac{1}{a}\operatorname{sh} at \sin x\right)$.

Т III. 4.

б) $u_{tt} = \Delta u + (x^2+y^2)\sin t$, $t > 0$, $(x,y,z) \in \mathbb{R}^3$; $u|_{t=0} = 0$, $u_t|_{t=0} = (x^2+y^2+z^2)^{5/2} - y^2$, $(x,y,z) \in \mathbb{R}^3$.

Частное решение: $u_2 = (Ax^2 + By^2 + C)\sin t \Rightarrow -Ax^2 - By^2 - C = 2A + 2B + x^2 + y^2 \Rightarrow$ $u_2 = (-3x^2 - 3y^2 + 4)\sin t$. $v := u - u_2 \Rightarrow v_{tt} = \Delta v$; $v|_{t=0} = u|_{t=0}$; $v_t|_{t=0} = (x^2+y^2+z^2)^{5/2} - 4 + x^2$.

1). $v|_{t=0} = 0$, $v_t|_{t=0} = (x^2+y^2+z^2)^{5/2}$. Ищем решение в виде $v(x,y,z,t) = w(r,t)$, $r = \sqrt{x^2+y^2+z^2} \Rightarrow (rw)_{tt} = (rw)_{rr} \Rightarrow s := rw \Rightarrow s_{tt} = s_{rr}$, $s|_{t=0} = 0$, $s_t|_{t=0} = r^6$, $r > 0$, $t > 0$ – задача на полуоси; доопределим нечётно для $r < 0$: $s_t|_{t=0} = -r^6 \Rightarrow$ ~~$s(t,r) = \frac{1}{2}\int_{r-t}^{r+t} \dots$~~

~~$s(r-t)$~~ $s(r,t) = F(r-t) + G(r+t)$, $r > t \Rightarrow F(r) + G(r) = 0$;

$-F'(r) + G'(r) = r^6 \Rightarrow G(r) = \frac{r^7}{14} + C$; $F(r) = -\frac{r^7}{14} - C$, $r > 0$.

при $r > t$: $s(r,t) = -\frac{(r-t)^7}{14} + \frac{(r+t)^7}{14}$. Непрерывность: $s(0,t) = F(-t) + G(t) = F(-t) + \frac{t^7}{14} + C = 0 \Rightarrow F(\xi) = -C + \frac{\xi^7}{14}$, $\xi < 0 \Rightarrow$ при $r < t$ ↳

$S(r,t) = \frac{(r+t)^7}{14} + \frac{(r+t)^7}{14} + C - C$ ~~(r+t)^7~~

2). $v_{tt} = \Delta v$; $v|_{t=0} = 0$; $v_t|_{t=0} = x^2 - 4 \Rightarrow v(x,t) = \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!} \Delta^n (x^2-4)$.

$\Delta^0 = x^2-4$; $\Delta = 2$; $\Delta^n = 0$ при $n \geq 2 \Rightarrow v(x,t) = t(x^2-4) + \frac{t^3}{3} \Rightarrow$

Ответ: $u(x,y,z,t) = (-x^2-y^2+4)\sin t + \frac{1}{14}\left(t+\sqrt{x^2+y^2+z^2}\right)^7 \pm \frac{1}{14}\left(\sqrt{x^2+y^2+z^2} - t\right)^7 + t(x^2-4) + \frac{t^3}{3}$ $\left("+" - t \geq \sqrt{x^2+y^2+z^2};\ "-" - t \leq \sqrt{x^2+y^2+z^2}\right]$

Ф. $u_{tt} = 8\Delta u + x$, $t>0$, $(x,y,z) \in \mathbb{R}^3$, $u|_{t=0} = e^{y-z}$; $u_t|_{t=0} = x^2+y^2+z^2$.

$u_2 = \frac{t^2}{2}x$, $v = u_0 = u - u_2 \Rightarrow v_{tt} = 8\Delta v$, $v|_{t=0} = e^{y-z}$, $v_t|_{t=0} = x^2+y^2+z^2$.

1). $v|_{t=0} = e^{y-z}$, $v_t|_{t=0} = 0 \Rightarrow$ ищем $v = w(\xi,t)$, $\xi = y - z \Rightarrow w_{tt} = 16 w_{\xi\xi}$,

$w(\xi,0) = e^{\xi} \Rightarrow w(\xi,t) = \frac{1}{2}\left(e^{\xi-4t} + e^{\xi+4t}\right) \Rightarrow v = e^{y-z}\operatorname{ch} 4t$.

2). $v|_{t=0} = 0$, $v_t|_{t=0} = x^2+y^2+z^2 \Rightarrow$ ищем $v(x,y,z,t) = \sum_{n=0}^{\infty} \frac{a^{2n}t^{2n}}{(2n)!}\Delta^n \varphi + \sum_{n=0}^{\infty} \frac{a^{2n}t^{2n+1}}{(2n+1)!}\Delta^n \psi = t(x^2+y^2+z^2) + 8t^3 \Rightarrow$ Ответ: $u(x,y,z,t) = \frac{t^2}{2}x + 8t^3 + t(x^2+y^2+z^2) + e^{y-z}\operatorname{ch} 4t$.

1). $u_{tt} = \Delta u - 81(t+1)^2 \operatorname{sh}(2x-2y+z)$, $t>0$, $(x,y,z) \in \mathbb{R}^3$; $u|_{t=0} = (2x+y-z)\cos y$,

$u_t|_{t=0} = x^2 - y^2 + z^2$. 1). $u_{tt} = \Delta u - 81(t+1)^2 \operatorname{sh}(2x-2y+z)$, $u|_{t=0} = 0$, $u_t|_{t=0} = 0$.

$u_2 = (At^2+Bt+C)\operatorname{sh}(2x-2y+z) \Rightarrow 2A\operatorname{sh}(2x-2y+z) = (At^2+Bt+C)(4+4+1) \cdot \operatorname{sh}(2x-2y+z) - 81(t+1)^2\operatorname{sh}(2x-2y+z) \Rightarrow u_2 = (9t^2+18t+11)\operatorname{sh}(2x-2y+z)$.

Ответ: ищем $u(x,y,z,t) = T_0(t)\operatorname{sh}(2x-2y+z) \Rightarrow T_0''(t)\operatorname{sh}(2x-2y+z) = 9T_0(t) \cdot \operatorname{sh}(2x-2y+z) + 81\operatorname{sh} \Rightarrow T_0(t) = C_1 e^{3t} + C_2 e^{-3t}$. Нач. усл.: $\begin{cases} C_1 + C_2 + 11 = 0 \\ 3C_1 - 3C_2 + 18 = 0 \end{cases} \Rightarrow$

$C_1 = \frac{17}{2}$, $C_2 = \frac{5}{2} \Rightarrow u(x,y,z,t) = (9t^2+18t+11)\operatorname{sh}(2x-2y+z) + \left(\frac{17}{2}e^{3t} + \frac{5}{2}e^{-3t}\right)\operatorname{sh}(2x-2y+z)$

2). $u_{tt} = \Delta u$; $u|_{t=0} = (2x+y-z)\cos y$, $u_t|_{t=0} = 0$. Ищем $u(x,y,z,t) = P(t) \cdot (2x+y-z)\cos y \Rightarrow P''(t)(2x+y-z)\cos y = -P(t)\sin y - P(t)\sin y - P(t)(2x+y-z) \cdot \cos y$ — не сходится $\Rightarrow$ ищем $v(x,y,z,t) = P(t)(2x+y-z)\cos y + S(t)\sin y \Rightarrow$

$P''(t)(2x+y-z)\cos y + S''(t)\sin y = -2P(t)\sin y - P(t)(2x+y-z)\cos y - S(t)\sin y \Rightarrow$

$S''(t) = -2P(t) - S(t)$; $P''(t) = -P(t) \Rightarrow P(t) = C_1\sin t + C_2\cos t$. Н.у.: $S(0) = 0$,

$P(0)=1$; $P'(0)=S'(0)=0 \Rightarrow P(t)=\cos t$; $S''(t)+S(t)=-2\cos t$ – резонанс $\Rightarrow$
$S(t)=-t\sin t \Rightarrow u(x,y,z,t)=\cos t\cos y(2x+y-z)-t\sin t\sin y$.

3). $u_{tt}=\Delta u$, $u|_{t=0}=0$, $u_t|_{t=0}=x^2-y^2+z^2 \Rightarrow$ 4-ф рвд получаем $u=t(x^2-y^2+z^2)+\frac{t^3}{3} \Rightarrow$
Ответ: $u(x,y,z,t)=t(x^2-y^2+z^2)+\frac{t^3}{3}+\cos t\cos y(2x+y-z)-t\sin t\sin y+$
$+\operatorname{sh}(2x-2y+z)\left(9t^2+18t+11+\frac{17}{2}e^{3t}+\frac{5}{2}e^{-3t}\right)$.

g) $u_{tt}=\frac{1}{5}\Delta u+2t^2\cos(x+2y)$, $t>0$, $(x,y,z)\in\mathbb{R}^3$; $u|_{t=0}=yz^3$, $u_t|_{t=0}=\frac{1}{1+(x-2z)^2}$.

Частное решение: $u_2=(At^2+B)\cos(x+2y) \Rightarrow 2A=-\frac{1}{5}(1+4)(At^2+B)+2t^2 \Rightarrow$
$A=2$, $B=-4 \Rightarrow u_2=(2t^2-4)\cos(x+2y)$. $v:=u-u_2$, $v_{tt}=\frac{1}{5}\Delta v$;
$v|_{t=0}=yz^3+4\cos(x+2y)$, $v_t|_{t=0}=\frac{1}{1+(x-2z)^2}$. 1). $v|_{t=0}=yz^3$, $v_t|_{t=0}=0 \Rightarrow$
4-3 рвд $\hookrightarrow v(x,y,z,t)=yz^3+\frac{3}{5}t^2yz$. 2). $v|_{t=0}=4\cos(x+2y)$, $v_t|_{t=0}=0$.
Ищем $v=P(t)\cdot\cos(x+2y) \Rightarrow P''(t)\cos(x+2y)=-P(t)\cos(x+2y) \Rightarrow P''(t)+P(t)=0 \Rightarrow$
$P(t)=C_1\cos t+C_2\sin t$. Н.у.: $P(0)=4$; $P'(0)=0 \Rightarrow C_1=4$, $C_2=0 \Rightarrow v=4\cos t\cos(x+2y)$.
3). $v|_{t=0}=0$, $v_t|_{t=0}=\frac{1}{1+(x-2z)^2}$ $\xi:=x-2z$, $v=:w(\xi,t)$; $w_{xx}=w_{\xi\xi}$, $w_{zz}=4w_{\xi\xi} \Rightarrow$
$w_{tt}=w_{\xi\xi}$, $w|_{t=0}=0$, $w_t|_{t=0}=\frac{1}{1+\xi^2} \Rightarrow w(\xi,t)=\frac{1}{2}\int_{\xi-t}^{\xi+t}\frac{1}{1+\eta^2}d\eta=\frac{1}{2}(\operatorname{arctg}(\xi+t)-$
$-\operatorname{arctg}(\xi-t)) \Rightarrow v=\frac{1}{2}(\operatorname{arctg}(x-2z+t)-\operatorname{arctg}(x-2z-t)) \Rightarrow$ Ответ: $u(x,y,z,t)=$
$=(2t^2-4+4\cos t)\cos(x+2y)+yz^3+\frac{3}{5}t^2yz+\frac{1}{2}(\operatorname{arctg}(x-2z+t)-\operatorname{arctg}(x-2z-t))$.

$p(t) = C_1\cos t + C_2\sin t$. Н.у.: $P(0)=4$; $P'(0)=0 \Rightarrow C_1=4$, $C_2=0 \Rightarrow v = 4\cos t\cos(x+2y)$.

3). $U|_{t=0}=0$, $U_t|_{t=0} = \frac{1}{1+(x-2z)^2}$ $\xi := x-2z$, $U =: W(\xi,t)$; $W_{xx}=W_{\xi\xi}$, $W_{zz}=4W_{\xi\xi} \Rightarrow$

$W_{tt}=W_{\xi\xi}$, $W|_{t=0}=0$, $W_t|_{t=0}=\frac{1}{1+\xi^2} \Rightarrow W(\xi,t)=\frac{1}{2}\int_{\xi-t}^{\xi+t}\frac{1}{1+\eta^2}d\eta = \frac{1}{2}(\operatorname{arctg}(\xi+t) -$

$-\operatorname{arctg}(\xi-t)) \Rightarrow U = \frac{1}{2}(\operatorname{arctg}(x-2z+t)-\operatorname{arctg}(x-2z-t)) \Rightarrow$ Ответ: $u(x,y,z,t) =$

$= (2t^2-4+4\cos t)\cos(x+2y) + yz^3 + \frac{3}{5}t^2yz + \frac{1}{2}(\operatorname{arctg}(x-2z+t)-\operatorname{arctg}(x-2z-t))$.

Т5.

$u_{tt}=a^2\Delta u$, $t>0$, $x=(x_1,x_2,x_3)\in\mathbb{R}^3$; $u|_{t=0}=\alpha(r)$, $u_t|_{t=0}=\beta(r)\in C^2(\mathbb{R})$, $r=\sqrt{x_1^2+x_2^2+x_3^2}$, $\alpha\in C^3(\mathbb{R})$

$u(x_1,x_2,x_3,t) =: W(r,t) \Rightarrow (rW)_{tt}=a^2(rW)_{rr}$. $V(r,t) := rW(r,t) \Rightarrow V_{tt}=a^2V_{rr}$,

$V|_{t=0}=r\alpha(r)$, $V_t|_{t=0}=r\beta(r)$; $r>0 \Rightarrow$ в соответствии с №212 продолжим нач. усл. нечётно на $r<0$: $V|_{t=0}=-r\alpha(-r)$, $V_t|_{t=0}=-r\beta(-r) \Rightarrow$ по ф-ле

Даламбера: $V(r,t)=\frac{1}{2}((r+at)\alpha(r+at)+(r-at)\alpha(r-at)) + \frac{1}{2a}\int_{r-at}^{r+at}\xi\beta(\xi)d\xi \Rightarrow$

$$u = W(r,t) = \begin{cases} \frac{1}{2r}((r+at)\alpha(r+at)+(r-at)\alpha(r-at)) + \frac{1}{2ar}\int_{r-at}^{r+at}\xi\beta(\xi)d\xi, & r\geq at \\ \frac{1}{2r}((r+at)\alpha(r+at)-(at-r)\alpha(at-r)) + \frac{1}{2ar}\int_{at-r}^{at+r}\xi\beta(\xi)d\xi, & r<at. \end{cases}$$

Т6.

$4u_{tt}=\Delta u$, $t>0$, $(x,y,z)\in\mathbb{R}^3$; $u|_{t=0}=0$, $u_t|_{t=0}=\begin{cases}\ln(1+x^2+y^2+z^2), & (x,y,z)\in G \\ 0, & (x,y,z)\in\mathbb{R}^3\setminus G,\end{cases}$

$G = \{y>0,\ 0<x<z\}$. $u(t,0,0,0)$ – ? По ф-ле Кирхгофа $\hookrightarrow u(0,t) = \frac{1}{4\pi\frac{1}{4}t}\iint_{|\xi|=\frac{t}{2}} u_t|_{t=0}(\xi)\,dS = \frac{1}{\pi t}\iint_{(|\xi|=\frac{1}{2})\cap G} \ln(1+\xi^2)\,dS$. На сф-сти $|\xi|=\frac{1}{2} \hookrightarrow \ln(1+\xi^2) = \ln\left(1+\frac{t^2}{4}\right) = \text{const} \Rightarrow u(0,t) = \frac{\ln\left(1+\frac{t^2}{4}\right)}{\pi t}\cdot S\left((|\xi|=\frac{1}{2})\cap G\right)$ $\Rightarrow$ перейдём к сферич. координ-м на сфере с $R=\frac{t}{2}$: $\begin{cases} x = R\sin\theta\cos\varphi \\ y = R\sin\theta\sin\varphi \\ z = R\cos\theta \end{cases}$, $\theta\in[0,\pi]$, $\varphi\in[0,2\pi)$. $\Rightarrow$

$G:\begin{cases}\sin\theta\sin\varphi>0\\ \sin\theta\cos\varphi>0\\ \sin\theta\cos\varphi<\cos\theta\end{cases} \Rightarrow \varphi\in\left(0;\frac{\pi}{2}\right),\ \theta\in\left(0,\operatorname{arctg}\frac{1}{\cos\varphi}\right) \Rightarrow$ площадь $S =$

$= R^2\int_0^{\pi/2}\int_0^{\operatorname{arctg}(1/\cos\varphi)}\sin\theta\,d\theta\,d\varphi = \frac{t^2}{4}\int_0^{\frac{\pi}{2}}\left(1-\cos\left(\operatorname{arctg}\frac{1}{\cos\varphi}\right)\right)d\varphi = \frac{t^2}{4}\int_0^{\frac{\pi}{2}}\left(1-\frac{\cos\varphi}{\sqrt{2+\cos^2\varphi}}\right)d\varphi =$

$= \frac{t^2}{4}\left(\frac{\pi}{2} - \int_0^1\frac{du}{\sqrt{2-u^2}}\right) = \frac{t^2}{4}\left(\frac{\pi}{2} - \arcsin\left(\frac{u}{\sqrt{2}}\right)\Big|_0^1\right) = \frac{t^2}{4}\left(\frac{\pi}{2}-\frac{\pi}{4}\right) = \frac{\pi t^2}{16} \Rightarrow$

$u(0,t) = \frac{\ln\left(1+\frac{t^2}{4}\right)}{\pi t}\cdot\frac{\pi t^2}{16} = \frac{t}{16}\ln\left(1+\frac{t^2}{4}\right)$.